Semester VII **B. Tech CSE (AI)**

BCSAI 702: SUPERVISED AND UNSUPERVISED LEARNING

3L+0T+0P+3 **MM:100**

Unit 1: Overview

Classification, Clustering, Constraints, One-pass learn concepts, Discard after learn concept

Unit 2: Supervised Learning

Logistic Regression, PERCEPTRONS, Neural Networks, Gaussian processes, Support Vector Machines

Unit 3: UnSupervised Learning

Identification, characterization and modeling ,clustering structure, quantum computing, cognitive techniques

Unit 4: Reinforced Learning

Environment, State, reward,policy, multi-task agents,state abstractions

Unit 5: Deep Reinforcement Learning

Continuity analysis,Monte Carlo Tree search , Gradient methods

Text/Reference Books

1. Machine Learning Algorithms for Supervised and Unsupervised Learning: The Future is Here by William Sullivan,Create space independent publishing platform, 2018

2. A systematic Review on Supervised and Unsupervised Machine Learning Algorithms for Data Science, Alloghani,Mohamed(et al.)

3. The elements of Statistical Learning,Trevor Hastie,Robert Tibshirani,Jerome Friedman; Springer